Heaven Alakraa

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EDUCATION

Brown University, Sc.B in Mechanical Engineering, GPA: 3.96/4.00 **Providence, RI | Aug 2024 - May 2028**
Relevant Coursework: Mechanics of Structures, Thermodynamics, Fluid Mechanics, Control Systems Engineering
Awards: 2026 Carlo De Luca Biomechanics Summer Undergraduate Research Experience (B-SURE) Award, American Society of Biomechanics

WORK EXPERIENCE

Engineering Research Intern | Bioengineering Laboratory **Providence, RI | May 2026 – Present**

- Designed a pendulum-based testing apparatus (SolidWorks 3D assembly, 2D drawing package, BOM) to measure the coefficient of friction of porcine wrist joints under body-weight loads of 18–60 kg through a ±15° swing range
- Fabricated and built the pendulum system using a drill press, hand drill, tap set, and powered saw, and verified vertical alignment with a level to establish a controlled, repeatable swing plane
- Instrumented the pendulum with an IMU and applied an exponential swing-decay method, fitting the angular-decay curve in MATLAB (lsqcurvefit) to extract the joint's coefficient of friction

Product Design Intern | Lilac Biosciences Inc. **Providence, RI | Jan 2026 – Present**

- Designed and fabricated 2 3D-printed prototype iterations of an RNA purification column that directly applies an electric current to extract RNA directly out of solution to replace the multi-step centrifuge-based spin-column
- Specified <0.2 µm pore size and >1 mm thickness criteria for a 100 µL sample volume to balance RNA retention, and developed the internal flow path and membrane-seat geometry in SolidWorks

Cooling and Testing Engineer | Brown Formula Racing Team (FSAE), **Providence, RI | Sep 2025 – Present**

- Designed and built a wind-tunnel test rig in SolidWorks to measure radiator airflow and thermal performance, sourcing piping and fluid-system components to support fabrication
- Measured radiator heat-transfer coefficients through 3-bucket calorimetry and mass-flow-rate testing across a range of operating conditions, generating experimental performance data for the cooling subteam

PROJECTS

Self-Balancing Robot

- Designed, wired, and programmed a two-wheeled self-balancing robot using Arduino Uno, MPU6050 IMU, L298N motor driver, and hand-tuned PID controller to hold upright balance without external support
- Diagnosed sensor drift and motor-direction faults through iterative bench testing; reduced steady-state lean via integral-term tuning and improved disturbance recovery through derivative damping
- Modeled and 3D printed a two-tier chassis in SolidWorks, integrating sensing, actuation, and power subsystems
- Achieved stable unsupported balance for 20 seconds with recovery from 20° manual disturbances

Can Opener: Mechanical Design, Documentation & Analysis

- Modeled a fully constrained 10-component rotary can-opener assembly in SolidWorks and produced a 6-sheet 2D drawings (ASME Y14.5-2009, Critical-to-Function dimensioning) with section, detail, and exploded views
- Ran static FEA under a 200 N grip load in AISI 304 stainless steel, confirming stresses remained well below the 207 MPa yield strength and identifying a stress-concentration artifact at an idealized boundary condition

Cantilever Beam Deflection Scale

- Designed and fabricated a cantilever-beam mechanical scale, applying Euler–Bernoulli beam theory and Hooke's law to translate analytical deflection equations into a physical build using a bandsaw and hand drill
- Optimized beam geometry and load placement in MATLAB, computing spring constant, bending stress, and safety factor, and validated the model through incremental physical calibration, achieving under 10% error

Solar Thermal Building Heating System

- Designed a 23-unit evacuated-tube solar collector array to offset 87,600 kWh/year of a building's heating demand
- Specified the circulator pump and glycol/water storage tank by calculating required flow rate, pressure head, and thermal storage capacity, predicted a ~7-year return on investment through a techno-economic study

SKILLS

CAD: SolidWorks, Fusion 360, Onshape, 2D Engineering Drawings, GD&T ASME Y14.5, Design for Manufacturing
Fabrication: Machining, Drill press, Hand tools, Tapping, Mill, Laser cutter, Band Saw, 3D printing
Simulation & Analysis: FEA, Abaqus, Motion Analysis, CAM Analysis, Computational Fluid Dynamics (CFD)
Programming & Data Analysis: MATLAB, Arduino, ImageJ, Python (Pandas, NumPy, PyCharm), LTspice, Simulink